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Shielding port for nuclear logging tool with gamma ray source

Abstract

A system includes a shield of a nuclear logging tool. The shield includes a shielding insert configured to mount at least partially inside the nuclear logging tool between a gamma ray source of the nuclear logging tool and a gamma ray detector of the nuclear logging tool. A first portion of the shielding insert is configured to mount in a collar of the nuclear logging tool and a second portion of the shielding insert is configured to mount in a chassis of the nuclear logging tool. The shield also includes a chassis shielding block configured to mount in the chassis between the gamma ray source and the gamma ray detector and a shielding top plate configured to couple to the collar and at least partially retain the shielding insert in the collar.

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Background/Summary

BACKGROUND

(1) The present disclosure relates generally to well logging techniques and, more particularly, to gamma ray density measurement for subterranean formations using a nuclear logging tool with a gamma ray source.

(2) This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the present techniques, which are described and/or claimed below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. Accordingly, it should be understood that these statements are to be read in this light, and not as admissions of prior art.

(3) Formation density is a crucial property of rocks for evaluating the hydrocarbon deposits. This property is usually determined by logging while drilling (LWD) during the drilling of a borehole through the formation in question, or wireline tools after the borehole has been drilled. The nuclear logging tool that is used for these measurements usually contains a radioisotopic source (e.g., .sup.137Cs or .sup.241AmBe) of high energy photons (e.g., gamma rays) and radiation detectors (e.g., gamma ray detectors). The formation density measurement involves the scattering of gamma

rays through the formation. A formation density may be obtained by irradiating the formation with gamma rays using the radioisotopic source (e.g., .sup.137Cs or .sup.241 AmBe). These gamma rays may Compton scatter from the electrons present in the formation before being detected by a gamma ray detector spaced some distance from the gamma ray source. Since the electron concentration is proportional to the atomic number of the elements, and the degree to which the gamma rays Compton scatter and return to the gamma ray detector relates to the electron concentration, the density of the formation may be measured using this technique.

(4) However, the gamma ray detector in the nuclear logging tool may detect gamma rays that transmit in the nuclear logging tool itself. For example, some gamma rays emitted from the gamma ray source may transmit inside the nuclear logging tool, such as a chassis of the logging tool, a collar of the logging tool, or other tool structures. The gamma rays detected outside of the formation may not provide substantial information regarding the properties of the formation and may represent noise that would be desired to reduce or subtracted from the overall signal received by the gamma ray detector. Accordingly, a need exists for shielding in the nuclear logging tool, wherein the shielding blocks undesirable gamma ray transmissions inside the nuclear logging tool.

SUMMARY

(5) A summary of certain embodiments disclosed herein is set forth below. It should be understood that these aspects are presented merely to provide the reader with a brief summary of these certain embodiments and that these aspects are not intended to limit the scope of this disclosure. Indeed, this disclosure may encompass a variety of aspects that may not be set forth below.

(6) Certain embodiments of the present disclosure include a system that includes a shield of a nuclear logging tool. The shield includes a shielding insert configured to mount at least partially inside the nuclear logging tool between a gamma ray source of the nuclear logging tool and a gamma ray detector of the nuclear logging tool. A first portion of the shielding insert is configured to mount in a collar of the nuclear logging tool and a second portion of the shielding insert is configured to mount in a chassis of the nuclear logging tool. The shield also includes a chassis shielding block configured to mount in the chassis between the gamma ray source and the gamma ray detector and a shielding top plate configured to couple to the collar and at least partially retain the shielding insert in the collar.

(7) Certain embodiments of the present disclosure include a nuclear logging tool. The nuclear logging tool includes a chassis, a collar disposed about the chassis, a gamma ray source configured to emit gamma rays through a first window in the collar, and a gamma ray detector configured to receive gamma rays through a second window in the collar, and the gamma ray source and the gamma ray detector are offset from one another along an axis of the chassis. The nuclear logging tool also includes a shield. The shield includes a shielding insert mounted at least partially inside the nuclear logging tool between the gamma ray source and the gamma ray detector. A first portion of the shielding insert is mounted in the collar and a second portion of the shielding insert is mounted in the chassis. The shield also includes a chassis shielding block mounted in the chassis between the gamma ray source and the gamma ray detector. The shield also includes a shielding top plate coupled to the collar and at least partially retaining the shielding insert in the collar.

(8) Certain embodiments of the present disclosure also include a method that includes emitting gamma rays from a gamma ray source through a first window in a collar of a nuclear logging tool; receiving gamma rays at a gamma ray detector through a second window in the collar, and the gamma ray source and the gamma ray detector are offset from one another along an axis of a chassis of the nuclear logging tool; and shielding gamma rays directly between the gamma ray source and the gamma ray detector internally within the nuclear logging tool via a shield. The shield includes a shielding insert mounted at least partially inside the nuclear logging tool between the gamma ray source and the gamma ray detector. A first portion of the shielding insert is mounted in the collar and a second portion of the shielding insert is mounted in the chassis. The shield also includes a chassis shielding block mounted in the chassis between the gamma ray source and the

gamma ray detector. The shield also includes a shielding top plate coupled to the collar and at least partially retaining the shielding insert in the collar.
(9) Various refinements of the features noted above may exist in relation to various aspects of the present disclosure. Further features may also be incorporated in these various aspects as well. These refinements and additional features may exist individually or in any combination. For instance, various features discussed below in relation to one or more of the illustrated embodiments may be incorporated into any of the above-described aspects of the present disclosure alone or in any combination. The brief summary presented above is intended only to familiarize the reader with certain aspects and contexts of embodiments of the present disclosure without limitation to the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Various aspects of this disclosure may be better understood upon reading the following detailed description and upon reference to the drawings in which:
(2) FIG. **1** is a block diagram of a drilling system for performing a gamma density measurement, in accordance with an embodiment;
(3) FIG. **2** is a schematic block diagram showing a perspective view of a nuclear logging tool used in the drilling system of FIG. **1**, in accordance with an embodiment;
(4) FIG. **3** is a partial perspective view of an illustrative embodiment of the nuclear logging tool used in the drilling system of FIG. **1**, in accordance with an embodiment;
(5) FIG. **4** is a partial cross-sectional end view of the nuclear logging tool used in the drilling system of FIG. **1**, in accordance with an embodiment;
(6) FIG. **5** is a partial cross-sectional side view of the nuclear logging tool used in the drilling system of FIG. **1**, in accordance with an embodiment;
(7) FIG. **6** is a partial cross-sectional side view of the nuclear logging tool used in the drilling system of FIG. **1** illustrating an embodiment of a shield of the nuclear logging tool, in accordance with an embodiment;
(8) FIG. **7** is a partial cross-sectional side view of the nuclear logging tool used in the drilling system of FIG. **1** illustrating a second embodiment of the shield of the nuclear logging tool, in accordance with an embodiment;
(9) FIG. **8** is a partial cross-sectional side view of the nuclear logging tool used in the drilling system of FIG. **1** illustrating a third embodiment of the shield of the nuclear logging tool, in accordance with an embodiment; and
(10) FIG. **9** is a partial cross-sectional side view of the nuclear logging tool used in the drilling system of FIG. **1** illustrating a fourth embodiment of the shield of the nuclear logging tool, in accordance with an embodiment.
DETAILED DESCRIPTION
(11) One or more specific embodiments of the present disclosure will be described below. These described embodiments are only examples of the presently disclosed techniques. Additionally, in an effort to provide a concise description of these embodiments, all features of an actual implementation may not be described in the specification. It should be appreciated that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be appreciated that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.

(12) When introducing elements of various embodiments of the present disclosure, the articles "a," "an," and "the" are intended to mean that there are one or more of the elements. The terms "comprising," "including," and "having" are intended to be inclusive and mean that there may be additional elements other than the listed elements. Additionally, it should be understood that references to "one embodiment" or "an embodiment" of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features.
(13) As used herein, the terms "connect," "connection," "connected," "in connection with," and "connecting" are used to mean "in direct connection with" or "in connection with via one or more elements"; and the term "set" is used to mean "one element" or "more than one element." Further, the terms "couple," "coupling," "coupled," "coupled together," and "coupled with" are used to mean "directly coupled together" or "coupled together via one or more elements."
(14) In addition, as used herein, the terms "real time", "real-time", or "substantially real time" may be used interchangeably and are intended to describe operations (e.g., computing operations) that are performed without any human-perceivable interruption between operations. For example, as used herein, data relating to the systems described herein may be collected, transmitted, and/or used in control computations in "substantially real time" such that data readings, data transfers, and/or data processing steps occur once every second, once every 0.1 second, once every 0.01 second, or even more frequent, during operations of the systems (e.g., while the systems are operating). In addition, as used herein, the terms "continuous", "continuously", or "continually" are intended to describe operations that are performed without any significant interruption. For example, as used herein, control commands may be transmitted to certain equipment every five minutes, every minute, every 30 seconds, every 15 seconds, every 10 seconds, every 5 seconds, or even more often, such that operating parameters of the equipment may be adjusted without any significant interruption to the closed-loop control of the equipment. In addition, as used herein, the terms "automatic", "automated", "autonomous", and so forth, are intended to describe operations that are performed are caused to be performed, for example, by a computing system (i.e., solely by the computing system, without human intervention). Indeed, although certain operations described herein may not be explicitly described as being performed continuously and/or automatically in substantially real time during operation of the computing system and/or equipment controlled by the computing system, it will be appreciated that these operations may, in fact, be performed continuously and/or automatically in substantially real time during operation of the computing system and/or equipment controlled by the computing system to improve the functionality of the computing system (e.g., by not requiring human intervention, thereby facilitating faster operational decision-making, as well as improving the accuracy of the operational decision-making by, for example, eliminating the potential for human error), as described in greater detail herein.
(15) Present embodiments relate to systems and techniques for using a nuclear logging tool with a gamma source and gamma detectors to get accurate formation density. For instance, a nuclear logging tool for obtaining such a measurement may include a gamma ray source, one or more gamma ray detectors, and data processing circuitry. When the nuclear logging tool is lowered into a borehole of a subterranean formation, the gamma ray source may irradiate gamma rays into the formation. The gamma rays interact with borehole and formation atoms and are attenuated by the formation before reaching the gamma detector(s). The data processing circuitry may determine the density of the formation based at least in part on the counts of gamma rays detected. To reduce or avoid the gamma rays transmit from the gamma ray source to the gamma ray detector(s) inside or through the nuclear logging tool, a shielding port may be used in the nuclear logging tool to sufficiently shield or block the gamma rays transmitting from the gamma ray source to the gamma ray detector(s) inside or through the nuclear logging tool, as described in greater detail herein.
(16) With the foregoing in mind, FIG. **1** illustrates a drilling system **10** having a nuclear logging tool, as described in greater detail herein. The drilling system **10** may be used to drill a well into a

geological formation **12** and obtain gamma ray spectroscopy measurements useful to identify characteristics of the well. In the drilling system **10** illustrated in FIG. **1**, a drilling rig **14** at the surface **16** may rotate a drill string **18** having a drill bit **20** at its lower end. As the drill bit **20** is rotated, a drilling fluid pump **22** is used to pump drilling fluid **24**, commonly referred to as "mud" or "drilling mud", downward through the center of the drill string **18** in the direction of the arrow to the drill bit **20**. The drilling fluid **24**, which is used to cool and lubricate the drill bit **20**, exits the drill string **18** through the drill bit **20**. The drilling fluid **24** then carries drill cuttings away from the bottom of a borehole **26** as it flows back to the surface **16**, as illustrated by the arrows through an annulus **30** between the drill string **18** and the formation **12**. However, as described above, as the drilling fluid **24** flows through the annulus **30** between the drill string **18** and the formation **12**, the drilling fluid **24** may begin to invade and mix with the fluids stored in the formation **12**, which may be referred to as formation fluid (e.g., natural gas or oil). At the surface **16**, the return drilling fluid **24** may be filtered and conveyed back to a mud pit **32** for reuse.

(17) As illustrated in FIG. **1**, the lower end of the drill string **18** includes a bottom-hole assembly (BHA) **34** that may include the drill bit **20** along with various downhole tools. The downhole tools may collect a variety of information relating to the geological formation **12** and/or the state of drilling of the well. For example, a measurement-while-drilling (MWD) module **36** may measure certain drilling parameters, such as the temperature, pressure, orientation of the drilling tool, and so forth. Likewise, a logging-while-drilling (LWD) module **38** may measure the physical properties of the geological formation **12**, such as density, porosity, resistivity, lithology, and so forth.

(18) The LWD module **38** may collect a variety of data **40** that may be stored and processed within the LWD module **38** or, as illustrated in FIG. **1**, may be sent to the surface for processing. In the example of this disclosure, the LWD module **38** may include a nuclear logging tool that may detect gamma ray spectroscopy (e.g., the energies of formation gamma rays) that result when gamma rays are emitted into the well. The range of energies of the detected gamma rays may be visualized as a spectrum of the gamma rays that are detected. The data **40** that is collected may include counts and/or detected energies of gamma rays and gamma rays that reach corresponding detectors in the LWD module **38**. It should be appreciated that while the embodiment illustrated in FIG. **1** is directed to collecting data via an LWD module **38**, in other embodiments, wireline tools may be used as the conveyance mode. In other words, the gamma ray nuclear logging tool may be deployed into the borehole **26** via LWD, wireline, coiled tubing, or any other suitable mode of downhole conveyance.

(19) The data **40** may be sent via a control and data acquisition system **42** to a data processing system **44**. The control and data acquisition system **42** may receive the data **40** in any suitable way. In certain embodiments, the control and data acquisition system **42** may transfer the data **40** via electrical signals pulsed through the geological formation **12** or via mud pulse telemetry using the drilling fluid **24**. In other embodiments, the data **40** may be retrieved directly from the LWD module **38** when the LWD module **38** returns to the surface. As described in greater detail herein, the control and data acquisition system **42** may be configured to estimate formation properties (e.g., porosity) using nuclear data **40** provided by the LWD module **38**. In addition, in certain embodiments, the control and data acquisition system **42** may be configured to control any and all operational parameters of the BHA **34** including, but not limited to, operations of a nuclear logging tool of the LWD module **38**, as described in greater detail herein.

(20) In certain embodiments, the data processing system **44** may include a processor **46**, memory **48**, storage **50**, and/or a display **52**. The data processing system **44** may use the data **40** to determine various properties of the formation **12** using any suitable techniques. As will be described in greater detail herein, the LWD module **38** may use certain selected materials to reduce signal contamination by stray gamma rays. Thus, when the data processing system **44** processes the data **40**, the determined formation properties may be more accurate and/or precise than otherwise. To process the data **40**, the processor **46** may execute instructions stored in the memory **48** and/or

storage **50**. As such, the memory **48** and/or the storage **50** of the data processing system **44** may be any suitable article of manufacture that can store the instructions. The memory **48** and/or the storage **50** may be ROM memory, random-access memory (RAM), flash memory, an optical storage medium, or a hard disk drive, to name a few examples. The display **52** may be any suitable electronic display that can display logs and/or other information relating to properties of the formation **12** as measured by the LWD module **38**. It should be appreciated that, although the data processing system **44** is illustrated as being located at the surface, the data processing system **44** may be located in the LWD module **38**. In such embodiments, some of the data **40** may be processed in the LWD module **38** and the data **40** may be stored in the LWD module **38**, while some of the data **40** may be sent to the surface in real time. This may be the case particularly in LWD, where a limited amount of the data **40** may be transmitted to the surface during drilling or reaming operations.

(21) As described in greater detail herein, the LWD module **38** may include a nuclear logging tool. FIG. **2** illustrates a schematic block diagram showing a perspective view of an embodiment of the LWD module **38** being or including a nuclear logging tool **54**. In certain embodiments, the nuclear logging tool **54** may include a chassis **56** (e.g., annular chassis), a collar **58** (e.g., annular collar), and a flow path **60** (e.g., annular flow path) that extends through the nuclear logging tool **54**. To facilitate discussion, the nuclear logging tool **54** and its components may be described with reference to an axial axis or direction **80** (e.g., a co-axis of the collar **58** and the chassis **56**), a radial axis or direction **82**, and a circumferential axis or direction **84**. In certain embodiments, a gamma ray source **62** may be located at a first location within the nuclear logging tool **54**, a first gamma ray detector **64** may be located at a second location axially spaced from the gamma ray source **62** along the axis **80**, a second gamma ray detector **66** may be located at a third location axially spaced from the gamma ray source **62** along the axis **80**, with the third location farther away from the gamma ray source **62** along the axis **80** than the second location. A shield **70** that includes dense shielding material (e.g., tungsten copper, HeavyMet made of tungsten nickel iron) may be disposed about the nuclear logging tool **54** at or near the second location to reduce the undesirable gamma rays traveling from the gamma ray source **62** directly to the gamma ray detectors **64** through the LWD tool. It should be noted that the shield **70** may also be disposed at or near the third location to reduce the undesirable gamma rays traveling from the gamma ray source **62** directly to the gamma ray detectors **66** through the LWD tool. The flow path **60** may be along the axis **80**, and a center symmetric axis **59** of the flow path **60** may be offset from a center symmetric axis **61** of the collar **58** along the radial axis **82**. Specifics regarding this embodiment and other embodiments of spectroscopic logging tools employing the general configuration or aspects of the LWD module **38** and the nuclear logging tool **54** are envisaged for use with any suitable means of conveyance, such as wireline, coiled tubing, logging while drilling (LWD), and so forth. Further, information regarding the environment, such as the sigma of the formation **12**, sigma of the drilling fluid **24**, density, borehole size, and slowdown length, may be acquired using additional equipment.

(22) In general, gamma rays emitted by the gamma ray source **62** may interact with the surrounding formation **12**. The formation gamma rays may be detected by the gamma ray detectors (e.g., the first gamma ray detector **64**, the second gamma ray detector **66**). In certain embodiments, the gamma ray detectors may include scintillation detectors having a scintillation crystal and a photomultiplier. In certain embodiments, the gamma ray detectors (e.g., the first gamma ray detector **64**, the second gamma ray detector **66**) may detect the spectra—that is, the range of energies—of the formation gamma rays. The nuclear spectroscopy provided by the nuclear logging tool **54** illustrated in FIG. **2** may provide a wealth of information on the elemental composition of materials around the nuclear logging tool **54**. As described in greater detail herein, the control and data acquisition system **42** may analyze the energy spectra to obtain complementary information from different sets of elements (e.g., elemental components of the surrounding formation **12**). The elemental yields from these spectra respond to the geochemistry of the formation rock as well as

solid organic matter, pore fluids, and the borehole **26**.

(23) As described in greater detail herein, the shield **70** may include dense shielding material (e.g., tungsten copper, HeavyMet made of tungsten nickel iron) to reduce undesirable gamma rays traveling from the gamma ray source **62** directly to the gamma ray detectors inside the nuclear logging tool **54**. FIG. **3** illustrates a partial perspective view of an illustrative embodiment of the nuclear logging tool **54** of FIGS. **1** and **2**, further illustrating the shield **70**. In the illustrated embodiment, the nuclear logging tool **54** includes a gamma ray source window **100** aligned with the gamma ray source **62**, a detector window **102** aligned with the gamma ray detector **64**, and the shield **70** disposed between the gamma ray source **62** and the gamma ray detector **64**. It should be noted that, although only the detector window **102** is shown in the embodiment illustrated in FIG. **3**, an individual detector window may be aligned with corresponding gamma ray detector (e.g., the gamma ray detector **64**, the gamma ray detector **66**). The shield **70** includes a chassis shielding block **104** coupled to (e.g., recessed into) the chassis **56** and a shielding assembly **106** coupled to (e.g., recessed into) the collar **58** in alignment with the chassis shielding block **104**. The shielding assembly **106** may include shielding top plate **108** (e.g., shielding cover), a sealing part **110** (e.g., annular sealing cap) having an annular seal **114** disposed in an annular seal groove **112**, a shielding insert **116** (e.g., annular shielding insert), and a shielding connection block **118** (e.g., curved shielding panel or semi-annular panel), which may be arranged one over another in a sequential arrangement and/or coaxial with one another in the radial direction **82**. Various aspects of the gamma ray source window **100**, the detector window **102**, and shield **70** are discussed in further detail below.

(24) In certain embodiments, the gamma ray source window **100** (e.g., structural window panel or plug) may be installed within a window recess or opening **120** on the outer side of the collar **58** at the location of the gamma ray source **62**, so that gamma rays generated by the gamma ray source **62** may be output to the formation **12** through the gamma ray source window **100**. The gamma ray source window **100** may be made of materials with low density and lower average atomic number (Z) (e.g., polyetheretherketone (PEEK), transparent or translucent materials) that allow transmitting of gamma rays in certain energy spectra range. The gamma ray source window **100** may be of any shapes suitable for covering the output gamma rays beam of the gamma ray source **62**. For example, the gamma ray source window **100** may be a truncated cylindrical structure **122** having the same symmetrical axis as the collar **58**. For example, the truncated cylindrical structure **122** may have a rectangular outer perimeter or border **124**, a flat or planar inner surface **126**, and a curved outer surface **128**. In certain embodiments, the curved outer surface **128** may be matched or contoured to an outer surface **130** (e.g., outer annular surface) of the collar **58**.

(25) In addition, the detector window **102** (e.g., structural window panel or plug) may be installed within a window recess or opening **132** on the outer side of the collar **58** at the location of the gamma ray detector **64**, so that the gamma ray detector **64** may receive gamma rays from the formation **12** through the detector window **102**. The detector window **102** may be made of materials with low density and lower average atomic number (Z) (e.g., polyetheretherketone (PEEK), transparent or translucent materials) that allow transmitting of gamma rays in the certain energy spectra range. The detector window **102** may be of any shapes suitable for receiving input gamma ray beam from the formation **12**. For example, in the illustrated embodiment, the detector window **102** may be a cylindrical structure **134** having a cylindrical sidewall **136**, a flat or planar inner surface **138**, and a curved outer surface **140**. In certain embodiments, the curved outer surface **140** may be matched to the outer surface **130** (e.g., outer annular surface) of the collar **58**.

(26) The chassis shielding block **104** (e.g., curved shielding panel or semi-annular panel) may be disposed in a recess **142** formed in the chassis **56** between the gamma ray source **62** and the gamma ray detector **64**. The chassis shielding block **104** may have a rectangular perimeter **144**, an inner surface **146** (e.g., curved or flat inner surface), and an outer surface **148** (e.g., curved or flat outer surface). In certain embodiments, the chassis shielding block **104** may have a thickness **150** (e.g., a

uniform thickness and/or a variable thickness) between the inner surface **146** and the outer surface **148**. For example, the inner and outer surfaces **146** and **148** may be parallel to one another to define the uniform thickness **150**. The chassis shielding block **104** may be made of dense shielding material (e.g., tungsten copper, HeavyMet made of tungsten nickel iron) and may be a relatively large block (e.g., comparing to the size of the gamma ray source **62**) having a size that fits in the recess **142** between the gamma ray source **62** and the gamma ray detector **64** to reduce undesirable gamma rays traveling from the gamma ray source **62** directly to the gamma ray detector **64** in the chassis **56**.

(27) As further illustrated in FIG. **3**, the shielding assembly **106** may be inserted into a shielding port **152** in the collar **58** and partially into a recess **154** in the chassis shielding block **104** between the gamma ray source **62** and the gamma ray detector **64**, thereby further blocking or shielding the transmission of gamma rays from the gamma ray source **62** to the gamma ray detector **64** through the collar **58**. The shielding assembly **106** may be disposed axially between the gamma ray source **62** and the gamma ray detector **64** in a straight line (e.g., axial line in the axial direction **80**), wherein the shielding assembly **106** radially overlaps a radial length **156** of the gamma ray source **62** in the radial direction **82**, and the shielding assembly **106** circumferentially overlaps the gamma ray source **62** and the gamma ray detector **64** in the circumferential direction **84**. In some embodiments, a shielding width or diameter **158** of the shielding assembly **106** is greater than a source width or diameter **160** of the gamma ray source **62**. As noted above, the shielding top plate **108**, the sealing part **110**, the shielding insert **116**, and the shielding connection block **118** may be arranged one over another in a sequential arrangement and/or coaxial with one another in the radial direction **82**. For example, the shielding top plate **108** may be disposed radially over and coupled to the sealing part **110** within the shielding port **152**, the sealing part **110** may be disposed radially over and coupled to the shielding insert **116** in the shielding port **152**, and the shielding insert **116** may be disposed radially over and coupled to the shielding connection block **118** disposed in the recess **154** in the chassis shielding block **104**.

(28) The components of the shielding assembly **106** may have a variety of shapes to facilitate insertion into and/or removal from the shielding port **152** and the recess **154** while the collar **58** is disposed about the chassis **56**, the gamma ray source **62**, and the gamma ray detector **64**. The shielding top plate **108** may have a truncated cylindrical structure **162** having the same symmetrical axis as the collar **58**. For example, the truncated cylindrical structure **162** may have a rectangular outer perimeter or border **164**, a flat or planar inner surface **166**, and a curved outer surface **168**. In certain embodiments, the curved outer surface **168** may be matched or contoured to the outer surface **130** (e.g., annular outer surface) of the collar **58**. However, in certain embodiments, the shielding top plate **108** may be an annular plug, a flat circular cover, or another suitable shape. In certain embodiment, the sealing part **110** and the shielding insert **116** may be annular components as shown in FIG. **3**, rectangular components, oval components, or other shapes. In certain embodiment, the shielding connection block **118** (e.g., curved shielding panel or semi-annular panel) may have a rectangular perimeter **170**, an inner surface **172** (e.g., curved or flat inner surface), and an outer surface **174** (e.g., curved or flat outer surface). In certain embodiments, the shielding connection block **118** may have a thickness **176** (e.g., a uniform thickness and/or a variable thickness) between the inner surface **172** and the outer surface **174**. For example, the inner and outer surfaces **172** and **174** may be parallel to one another to define the uniform thickness **176**. The inner surface **172** of the shielding connection block **118** also may be contoured to fit along the outer surface **148** (e.g., curved or flat outer surface) and/or within the recess **154** in the chassis shielding block **104**. The shielding connection block **118** may be used to connect the shielding insert **116** to the chassis shielding block **104**. The shielding top plate **108** may be used to retain the sealing part **110** and the shielding insert **116** to the shielding connection block **118**. The shielding top plate **108**, the shielding insert **116**, and the shielding connection block **118** may be made of dense shielding material (e.g., tungsten copper, HeavyMet made of tungsten nickel iron). In some

embodiments, the sealing part **110** may be made of metal (e.g., steel, stainless steel, Nickle based alloys, copper based alloys) for machining purpose, such as Monel, Inconel, MP35N, 17-4 PH, Beryllium copper. In some embodiments, the sealing part **110** may be made of dense shielding material.

(29) FIG. **4** is a partial cross-sectional end view of the nuclear logging tool **54** of FIGS. **1-3**, further illustrating details of the shield **70** having the shielding assembly **106** inserted within the shielding port **152** in the collar **58** and partially into the recess **154** in the chassis shielding block **104**. As illustrated in FIG. **4**, the collar **58** (e.g., annular collar) is disposed about the chassis **56** (e.g., annular chassis) in a coaxial or concentric arrangement, such that an inner surface **190** (e.g., inner annular surface) of the collar **58** faces and/or contacts an outer surface **192** (e.g., outer annular surface) of the chassis **56**. The chassis shielding block **104** is disposed in a recess **142** in the chassis **56**, wherein the recess **142** extends radially inwardly from the outer surface **192** of the chassis **56** into an interior portion **194** of the chassis **56**. In certain embodiments, the recess **142** in the chassis **56** is a rectangular slot having flat side surfaces **196** and a flat bottom surface **198** to fit with the rectangular perimeter **144** and the inner surface **146** of the chassis shielding block **104**. The chassis shielding block **104** may be sized substantially larger than the gamma ray source **62** and/or the gamma ray detector **64**. In the illustrated embodiment, a width **200** and a height **202** of the chassis shielding block **104** may be at least equal to or greater than 20, 30, 40, or 50 percent of a diameter **204** of the chassis **56**. Additionally, the width **200** and the height **202** of the chassis shielding block **104** may be at least equal to or greater than 1.1, 1.2, 1.3, 1.4, 1.5, 2, 3, or 4 times the diameter **160** of the gamma ray source **62**. Thus, the chassis shielding block **104** may be a relatively large block (e.g., comparing to the size of the gamma ray source **62**) locating in the chassis **56**. With reference to FIGS. **3**, **4**, and **5**, the gamma ray source **62** may be located at a position along the radial direction **82**, such that the gamma ray source **62** is radially separated or offset from the gamma ray detector **64**.

(30) As further illustrated in FIG. **4**, the shielding top plate **108** (e.g., shielding cover), the sealing part **110** (e.g., annular sealing cap), the shielding connection block **118** (e.g., curved shielding panel or semi-annular panel), and chassis shielding block **104** are arranged one over another in the sequential arrangement and/or coaxial with one another in the radial direction **82**. As discussed above, the shielding top plate **108** has the truncated cylindrical structure **162** with the flat or planar inner surface **166** disposed against the sealing part **110** and the curved outer surface **168** contoured along the outer surface **130** (e.g., annular outer surface) of the collar **58**. The shielding top plate **108** may be disposed in a first open portion **206** of the shielding port **152**, wherein the first open portion **206** may have a rectangular shape (e.g., rectangular recess) to receive and hold the shielding top plate **108**.

(31) The sealing part **110** may be disposed in a second open portion **208** of the shielding port **152**, wherein the second open portion **208** may include a first annular bore sized to receive and hold the sealing part **110**. For example, the annular seal **114** in the annular seal groove **112** of the sealing part **110** may be configured to seal along the second open portion **208** (e.g., first annular bore). The illustrated sealing part **110** (e.g., annular sealing cap) has a cup-shaped structure **210** with a circular top wall **212** coupled to an annular side wall **214**, wherein the cup-shaped structure **210** defines an inner annular chamber **216**.

(32) The shielding insert **116** includes an annular flange **218** disposed axially between a first annular portion **220** (e.g., annular head) and a second annular portion **222** (e.g., annular body). The first annular portion **220** is disposed in the inner annular chamber **216** of the sealing part **110**. The annular flange **218** abuts a bottom surface **224** of the annular side wall **214** of the sealing part **110** and a bottom annular ledge **226** of the second open portion **208** of the shielding port **152**. The second annular portion **222** extends through a third open portion **228** (e.g., second annular bore) of the shielding port **152**.

(33) At least some components of the shielding assembly **106** (e.g., shielding top plate **108**, sealing

part **110**, and shielding insert **116**) are configured to be radially inserted into the shielding port **152** using progressively smaller cross-sectional areas of the components and the shielding port **152** in a radial inward direction. As illustrated, the annular flange **218** has a greater diameter than the first and second annular portions **220** and **222** of the shielding insert **116**, and the first annular portion **220** has a greater diameter than the second annular portion **222** of the shield insert **116**. Additionally, the first open portion **206** (e.g., rectangular recess) has a greater size than a first diameter of the second open portion **208** (e.g., first annular bore) of the shielding port **152**, and the first diameter of the second open portion **208** (e.g., first annular bore) is greater than a second diameter of the third open portion **228** (e.g., second annular bore) of the shielding port **152**, such that the shielding port **152** gradually decreases in size (e.g., width or diameter) in the radially inward direction and gradually increases in size in the radially outward direction. This gradually changing size of the shielding port **152** facilitates insertion and removal of at least part of the shielding assembly **106** (e.g., shielding top plate **108**, sealing part **110**, and shielding insert **116**) from outside of the collar **58** of the nuclear logging tool **54**, whereas the shielding connection block **118** and the chassis shielding block **104** may be installed on the chassis **56** before and/or during assembly of the collar **58** around the chassis **56**. Additional details of the shielding assembly **106** are discussed in further detail below.

(34) FIG. **5** is a partial cross-sectional side view of the nuclear logging tool **54** of FIGS. **1-4**, further illustrating gamma ray paths outside of the nuclear logging tool **54** and partially inside of the nuclear logging tool **54** and blocked by the shield **70**. As illustrated in FIG. **5**, gamma rays may be generated by the gamma ray source **62** and transmit inside the collar **58**. Some gamma rays may travel along a path **230** inside the collar **58** and be blocked by the shielding assembly **106** (e.g., blocked by the shielding top plate **108**, or the shielding insert **116**). These gamma rays may be blocked by the dense shielding material of the shielding assembly **106** and may not be received by the gamma ray detector **64**. Some gamma rays may travel along a path **232** inside the collar **58** and be attenuated by the shielding assembly **106** (e.g., by the sealing part **110**). These gamma rays may be attenuated by the metal material of the shielding assembly **106**, and only a portion of these gamma rays may be received by the gamma ray detector **64**. Some gamma rays may travel through the gamma ray window **100** and exit to the formation **12**. These gamma rays may travel along a path **234**, which may extend to the formation **12**, and return to the detector window **102** and be received by the gamma ray detector **64**. Those gamma rays travelling along the path **230** and **232** may not provide substantial information regarding the properties of the formation and may represent noise that needs to be subtracted from the overall signal received by the gamma ray detector **64**. Those gamma rays travelling along the path **234** may provide information regarding the properties of the formation and may be included in the signal of the gamma ray detector **64**. Accordingly, by using the shielding assembly **106**, the gamma rays transmit from the gamma ray source **62** to the gamma ray detector **64** inside or through the nuclear logging tool (e.g., the chassis **56**, the collar **58**) may be sufficiently reduced.

(35) FIG. **6** is a partial cross-sectional side view of the nuclear logging tool **54** of FIGS. **1-5**, further illustrating an embodiment of the shield **70** having a threaded fastener assembly **240** coupling together the shielding insert **116** and the shielding connection block **118**. In the illustrated embodiment, the threaded fastener assembly **240** includes a male threaded fastener **242** (e.g., bolt) and a female threaded fastener **244** (e.g., nut). The male threaded fastener **242** includes a head **246**, a shaft **248** coupled to and radially extending from the head **246**, and a threaded portion **250** (e.g., male threads) along the shaft **248**. The head **246** may be a tool-engageable head having a tool interface **252**, such as a male or female tool interface having flats, slots, or other torque transfer features, configured to transfer torque from a tool to the male threaded fastener **242**. The female threaded fastener **244** includes a threaded bore **254** (e.g., female threads) extending through a fastener body **256**, such as a rectangular block or a hexagonal block. The fastener **256** may be made of a material different from the shielding connection block **118** (e.g., steel). The male threaded

fastener **242** is configured to thread into the female threaded fastener **244** to radially pull together the shielding insert **116** and the shielding connection block **118**, thereby securing the shielding insert **116** and the shielding connection block **118** within the nuclear logging tool **54**.
(36) In the illustrated embodiment, the shield **70** is assembled into the nuclear logging tool **54** partially before installation of the collar **58** around the chassis **56** and partially after installation of the collar **58** around the chassis **56**. For example, before installing the collar **58** around the chassis **56**, the chassis shielding block **104** may be installed into the recess **142** in the chassis **56**, the female threaded fastener **244** may be installed into a recess **258** in the inner surface **172** of the shielding connection block **118**, and the shielding connection block **118** (with the female threaded fastener **244**) may be installed into the recess **154** in the chassis shielding block **104**. The recess **258** and the female threaded fastener **244** may have complementary anti-rotation shapes that interface with one another (e.g., rectangular, hexagon, or other non-circular shapes), such that the female threaded fastener **244** does not rotate during rotation of the male threaded fastener **242** as discussed below. Initially, the shielding connection block **118** (with the female threaded fastener **244**) may be inserted in the radial inward direction **82** into the recess **154**, such that the inner surface **172** of the shielding connection block **118** contacts or abuts a bottom surface **260** of the recess **154** and the outer surface **174** of the shielding connection block **118** is flush with, recessed into, or slightly protruding from the outer surface **148** of the chassis shielding block **104**. The collar **58** may then be installed around the chassis **56**, the chassis shielding block **104**, and the shielding connection block **118**. The remaining components of the shield **70** (e.g., shielding assembly **106**) may be installed externally through the shielding port **152** in the collar **58** after the collar **58** has been installed around the chassis **56**.
(37) As further illustrated in FIG. **6**, the shielding insert **116** may be installed in the radial inward direction **82** into the shielding port **152** until the second annular portion **222** of the shielding insert **116** extends through the third open portion **228** (e.g., second annular bore) in the collar **58** and extends into a recess **262** in the top surface **174** of the shielding connection block **118**. The recess **262** and the second annular portion **222** may be sized and shaped to mate with one another, such as an annular interface. Additionally, the shielding insert **116** may be installed in the radial inward direction **82** until the annular flange **218** abuts the bottom annular ledge **226** of the second open portion **208** (e.g., an annular portion **264**) of the collar **58**. Subsequently, the male threaded fastener **242** may be inserted through a fastener receptacle **266** in the shielding insert **116**, through a fastener receptacle **268** in the shielding connection block **118**, and into the threaded bore **254** in the female threaded fastener **244**. A torque tool may then engage the head **246** (e.g., torque interface **252**) to rotate the male threaded fastener **242**, such that the threaded portion **250** of the shaft **248** threads into the threaded bore **254** in the female threaded fastener **244** and the head **246** recesses into a head recess **270** along an outer surface **272** (e.g., top surface) of the shielding insert **116**. As the torque tool gradually tightens the male threaded fastener **242** with the female threaded fastener **244**, the threaded fastener assembly **240** radially compresses the shielding insert **116** and the shielding connection block **118** radially about the annular portion **264** of the collar **58**. Accordingly, the tightening of the threaded fastener assembly **240** causes a radial inward force of the annular flange **218** of the shielding insert **116** against the bottom annular ledge **226** at the annular portion **264**, and a radial outward force of the outer surface **174** of the shielding connection block **118** against the inner surface **190** of the collar **58**. Thus, the threaded fastener assembly **240** removably couples the shielding insert **116** to the shielding connection block **118** while the collar **58** is mounted around the chassis **56**.
(38) The shielding assembly **106** also includes the sealing part **110** and the shielding top plate **108**. As illustrated in FIG. **6**, after installation of the male threaded fastener **266**, the sealing part **110** may be installed in the radial inward direction **82** into the second open portion **208** (e.g., first annular bore) of the shielding port **152** until the sealing part **110** substantially surrounds the first annular portion **220** of the shielding insert **116**. The sealing part **110** has the cup-shaped structure

210 defining the inner annular chamber **216**, which receives the first annular portion **220** of the shielding insert **116**. The circular top wall **212** of the cup-shaped structure **210** extends over and covers the male threaded fastener **242** having the head **246** disposed in the head recess **270**. The annular side wall **214** of the cup-shaped structure **210** also includes the annular seal **114** in the annular seal groove **112**, wherein the annular seal **114** seals against the second open portion **208** (e.g., first annular bore) of the shielding port **152** to block fluid leakage into the nuclear logging tool **54**. In some embodiments, the shielding assembly **106** may include one or more fasteners to couple the sealing part **110** to the collar **58**.
(39) In the illustrated embodiment, after installation of the sealing part **110**, the shielding top plate **108** may be installed in the radial inward direction **82** into the first open portion **206** over the sealing part **110**. The shielding top plate **108** may be removably coupled to the collar **58** via one or more fasteners **274**, such as threaded fasteners. For example, the fasteners **274** may couple the shielding top plate **108** to the collar **58** in a border portion **276** of the shielding top plate **108** outside of the perimeter of the sealing part **110**. However, any suitable arrangement of the fasteners **274** may be used to couple the shielding top plate **108** to the collar **58**. The shield **70** may be removed in a removal process having a reverse order relative to the installation process described in detail above. Additionally, the shield **70** may have various modifications in the placement and/or geometry of the components while still enabling the installation and removal processes, which are at least partially performed while the collar **58** is mounted around the chassis **56**.
(40) In the embodiment illustrated in FIG. **6**, a spring **278** may be inserted between the sealing part **110** and the shielding insert **116** (e.g., between the outer surface **272** and the sealing part **110**) for support or for compensation of the thermal expansion caused by parts made of different materials. For example, the spring may include a helical spring, a spring washer, an elastomeric insert, or any combination thereof.
(41) FIG. **7** is a partial cross-sectional side view of the nuclear logging tool **54** of FIGS. **1-6**, further illustrating a second embodiment of the shield **70** having the threaded fastener assembly **240** coupling together the shielding insert **116** and the shielding connection block **118**. The embodiment of FIG. **7** is substantially the same as the embodiment of FIG. **6**, except that the threaded fastener assembly **240** in the embodiment of FIG. **7** integrates the female threaded fastener **244** with shielding connection block **118** as a one-piece structure **280** (e.g., shielding connection block **118** with integrated female threaded fastener **244**). In other words, the one-piece structure **280** includes both the shielding connection block **118** and the female threaded fastener **244** as a single continuous block of material, wherein the threaded bore **254** (e.g., female threads) extends through the one-piece structure **280** rather than a separate fastener body **256**. In certain embodiments, the shielding connection block **118** in FIG. **7** may be made of metal (e.g., steel, stainless steel, Nickle based alloys, copper based alloys) while the shielding connection block **118** in the embodiment illustrated in FIG. **6** may be made of dense shielding material (e.g., tungsten copper, HeavyMet made of tungsten nickel iron). Thus, like elements are shown with like element numbers in FIGS. **6** and **7**. Unless stated otherwise, the elements of FIG. **7** are substantially the same as described above with reference to FIG. **6**.
(42) FIG. **8** is a partial cross-sectional side view of the nuclear logging tool **54** of FIGS. **1-6**, further illustrating a third embodiment of the shield **70** having the threaded fastener assembly **240** coupling together the shielding insert **116** and a first chassis portion **300** (e.g., a radially protruding annular portion) of the chassis **56** through an opening **302** (e.g., an annular opening) on the chassis shielding block **104**. The embodiment of FIG. **8** is substantially the same as the embodiment of FIG. **7**, except that the shielding insert **116** in the embodiment of FIG. **8** is coupled to the chassis **56** through the shielding block **104** without using the shielding connection block **118**. Thus, like elements are shown with like element numbers in FIGS. **6** and **8**. Unless stated otherwise, the elements are substantially the same as described above with reference to FIG. **6**. In FIG. **8**, the first chassis portion **300** may include a threaded bore **304** (e.g., female threads) extending through the

first chassis portion **300**. The male threaded fastener **242** is configured to thread into the threaded bore **304** of the first chassis portion **300** to radially pull together the shielding insert **116** and the chassis **56**, thereby securing the shielding insert **116** within the nuclear logging tool **54**. The first chassis portion **300** may have a width or diameter **306** and be supported by a second chassis portion **308** (e.g., a radially protruding annular portion) having a width or diameter **310**. In some embodiments, the width or diameter **310** may be larger than the width or diameter **306**.

(43) As illustrated, the first and second chassis portions **300** and **308** collectively define a stepped annular protrusion **312** extending radially outward from the chassis **58** into a stepped annular receptacle **314** in the chassis shielding block **104**. The stepped annular receptacle **314** includes a first receptacle **316** (e.g., annular receptacle) sized to receive the second annular portion **222** of the shielding insert **116** and the first chassis portion **300** of the stepped annular protrusion **312**. The stepped annular receptacle **314** also includes a second receptacle **318** (e.g., annular receptacle) sized to receive the second chassis portion **308** of the stepped annular protrusion **312**. In operation, as the male threaded fastener **242** engages the threaded bore **304** of the first chassis portion **300**, the shielding insert **116** is forced radially inward into the first receptacle **316** against the first chassis portion **300**, thereby removably securing the shielding insert **116** within the collar **58**.

(44) FIG. **9** is a partial cross-sectional side view of the nuclear logging tool **54** of FIGS. **1-6**, further illustrating a fourth embodiment of the shield **70** without the threaded fastener assembly **240** and the shielding connection block **118**. The embodiment of FIG. **9** is substantially the same as the embodiment of FIG. **6**, except that the shielding insert **116** in the embodiment of FIG. **9** is coupled to the shielding block **104** without using the threaded fastener assembly **240**. Thus, like elements are shown with like element numbers in FIGS. **6** and **8**. Unless stated otherwise, the elements are substantially the same as described above with reference to FIG. **6**. In FIG. **9**, the shielding insert **116** may extend radially into a recess **350** (e.g., annular recess) in the chassis shielding block **104**, wherein the shielding insert **116** is held in place by the sealing part **110** and the shielding top plate **108**. For example, the shielding top plate **108** may be secured to the collar **58** by the threaded fasteners **274**, a brazed joint, a welded joint, an adhesive joint, or a combination thereof, between the shielding top plate **108** and the collar **58**. In some embodiments, a brazed joint, a welded joint, and/or an adhesive joint may couple together the shielding insert **116** and the sealing part **110**, the sealing part **110** and the shielding top plate **108**, or a combination thereof. In some embodiments, a brazed joint, a welded joint, and/or an adhesive joint may couple together the collar **58** and one or more of the shielding insert **116**, the sealing part **110**, and the shielding top plate **108**.

(45) The technical effect of the disclosed embodiments is to provide shielding internally within the nuclear logging tool **54**, such as within the collar **58**, between the gamma ray source **62** and the gamma ray detector **64**, thereby reducing the possibility of undesirable gamma ray transmission within the nuclear logging tool **54** without passing the gamma rays through a formation outside of the nuclear logging tool **54**. The disclosed embodiments provide a shield **70** with a shielding assembly **106** coupled to (e.g., recessed into) the collar **58** in alignment with a chassis shielding block **104**. The shielding assembly **106** is at least partially insertable and removable while the collar **58** is mounted around the chassis **56**. For example, the shielding assembly **106** may include at least the shielding top plate **108**, the sealing part **110**, and the shielding insert **116**, which can be radially inserted or removed while the collar **58** is mounted around the chassis **56**. The shield **70** helps block the undesirable gamma ray transmissions to reduce noise, while allowing desirable gamma ray transmissions to pass through a formation outside of the nuclear logging tool **54** between the gamma ray source **62** and the gamma ray detector **64**.

(46) The subject matter described in detail above may be defined by one or more clauses, as set forth below.

(47) A system includes a shield of a nuclear logging tool. The shield includes a shielding insert configured to mount at least partially inside the nuclear logging tool between a gamma ray source of the nuclear logging tool and a gamma ray detector of the nuclear logging tool. A first portion of

the shielding insert is configured to mount in a collar of the nuclear logging tool and a second portion of the shielding insert is configured to mount in a chassis of the nuclear logging tool. The shield also includes a chassis shielding block configured to mount in the chassis between the gamma ray source and the gamma ray detector and a shielding top plate configured to couple to the collar and at least partially retain the shielding insert in the collar.

(48) The system of any preceding clause, wherein the shielding insert is made of dense shielding material.

(49) The system of any preceding clause, wherein the chassis shielding block is made of dense shielding material.

(50) The system of any preceding clause, including a sealing part coupled to the shielding insert and configured to seal the shielding insert in the collar.

(51) The system of any preceding clause, wherein the sealing part is brazed to the shielding insert.

(52) The system of any preceding clause wherein the first calculated formation density is used to compensate the second calculated formation density to obtain a compensated formation density.

(53) The system of any preceding clause, including a shielding connection block configured to couple the shielding insert to the chassis shielding block.

(54) The system of any preceding clause, wherein the shielding connection block is made of dense shielding material.

(55) The system of any preceding clause, including a threaded fastener configured to couple together the shielding insert and the shielding connection block.

(56) The system of any preceding clause, including a threaded fastener configured to couple together the shielding insert and the chassis shielding block.

(57) The system of any preceding clause, including the nuclear logging tool including the collar disposed about the chassis, wherein a shielding port extends radially through the collar toward the chassis, the shielding insert is disposed in the shielding port over the chassis shielding block, a sealing part is disposed in the shielding port over the shielding insert, and the shielding top plate is coupled to the collar over the sealing part.

(58) The system of any preceding clause, wherein the shielding port includes an annular sealing port having a first annular bore, a second annular bore, and an annular ledge between the first and second annular bores, wherein the sealing part includes an annular sealing cap having a cup-shaped structure disposed in the first annular bore, wherein the first portion of the shielding insert includes a first annular portion disposed in the cup-shaped structure, the second portion of the shielding insert includes a second annular portion disposed in the second annular bore, and an annular flange of the shielding insert is disposed between the cup-shaped structure and the annular ledge.

(59) The system of any preceding clause, wherein a first threaded fastener extends through the shielding insert and couples to a second threaded fastener in the shielding connection block, the chassis shielding block, or the chassis, wherein the first and second threaded fasteners are configured to thread together to hold the shielding insert in the shielding port.

(60) The system of any preceding clause, wherein the shielding top plate is coupled to the collar via one or more fasteners, a brazed joint, a welded joint, or an adhesive joint.

(61) A nuclear logging tool includes a chassis, a collar disposed about the chassis, a gamma ray source configured to emit gamma rays through a first window in the collar, and a gamma ray detector configured to receive gamma rays through a second window in the collar, and the gamma ray source and the gamma ray detector are offset from one another along an axis of the chassis. The nuclear logging tool also includes a shield. The shield includes a shielding insert mounted at least partially inside the nuclear logging tool between the gamma ray source and the gamma ray detector. A first portion of the shielding insert is mounted in the collar and a second portion of the shielding insert is mounted in the chassis. The shield also includes a chassis shielding block mounted in the chassis between the gamma ray source and the gamma ray detector. The shield also includes a shielding top plate coupled to the collar and at least partially retaining the shielding insert in the

collar.
(62) The nuclear logging tool of the preceding clause, wherein a shielding port extends radially through the collar toward the chassis, the shielding insert is disposed in the shielding port over the chassis shielding block, a sealing part is disposed in the shielding port over the shielding insert, and the shielding top plate is coupled to the collar over the sealing part.
(63) The nuclear logging tool of the preceding clause, wherein the shielding port includes an annular sealing port having a first annular bore, a second annular bore, and an annular ledge between the first and second annular bores, wherein the sealing part includes an annular sealing cap having a cup-shaped structure disposed in the first annular bore, wherein the first portion of the shielding insert includes a first annular portion disposed in the cup-shaped structure, the second portion of the shielding insert includes a second annular portion disposed in the second annular bore, and an annular flange of the shielding insert is disposed between the cup-shaped structure and the annular ledge.
(64) The nuclear logging tool of the preceding clause, including a shielding connection block configured to couple the shielding insert to the chassis shielding block, wherein a first threaded fastener extends through the shielding insert and couples to a second threaded fastener in the shielding connection block, the chassis shielding block, or the chassis, wherein the first and second threaded fasteners are configured to thread together to hold the shielding insert in the shielding port.
(65) A method includes emitting gamma rays from a gamma ray source through a first window in a collar of a nuclear logging tool; receiving gamma rays at a gamma ray detector through a second window in the collar, and the gamma ray source and the gamma ray detector are offset from one another along an axis of a chassis of the nuclear logging tool; and shielding gamma rays directly between the gamma ray source and the gamma ray detector internally within the nuclear logging tool via a shield. The shield includes a shielding insert mounted at least partially inside the nuclear logging tool between the gamma ray source and the gamma ray detector. A first portion of the shielding insert is mounted in the collar and a second portion of the shielding insert is mounted in the chassis. The shield also includes a chassis shielding block mounted in the chassis between the gamma ray source and the gamma ray detector. The shield also includes a shielding top plate coupled to the collar and at least partially retaining the shielding insert in the collar.
(66) The method of any preceding clause, wherein emitting the gamma rays includes directing the gamma rays through a geological formation, receiving the gamma rays includes receiving the gamma rays after passing through the geological formation, and shielding the gamma rays includes reducing noise associated with the gamma rays not passing through the geological formation.
(67) The specific embodiments described above have been illustrated by way of example, and it should be understood that these embodiments may be susceptible to various modifications and alternative forms. It should be further understood that the claims are not intended to be limited to the particular forms disclosed, but rather to cover all modifications, equivalents, and alternatives falling within the spirit and scope of this disclosure.
(68) In the claims, means-plus-function clauses are intended to cover the structures described herein as performing the recited function and not only structural equivalents, but also equivalent structures. Thus, for example, although a nail and a screw may not be structural equivalents in that a nail employs a cylindrical surface to secure wooden parts together, whereas a screw employs a helical surface, in the environment of fastening wooden parts, a nail and a screw may be equivalent structures. It is the express intention of the applicant not to invoke 35 U.S.C. § 112, paragraph 6 for any limitations of any of the claims herein, except for those in which the claim expressly uses the words “means for” together with an associated function.

Claims

1. A system, comprising: a shield of a nuclear logging tool, comprising: a shielding insert configured to mount at least partially inside the nuclear logging tool between a gamma ray source of the nuclear logging tool and a gamma ray detector of the nuclear logging tool, wherein a first portion of the shielding insert is configured to mount in a collar of the nuclear logging tool and a second portion of the shielding insert is configured to mount in a chassis of the nuclear logging tool; a chassis shielding block configured to mount in the chassis between the gamma ray source and the gamma ray detector; and a shielding top plate configured to couple to the collar and at least partially retain the shielding insert in the collar.
2. The system of claim 1, wherein the shielding insert is made of dense shielding material.
3. The system of claim 2, wherein the chassis shielding block is made of dense shielding material.
4. The system of claim 1, further comprising a sealing part coupled to the shielding insert and configured to seal the shielding insert in the collar.
5. The system of claim 4, wherein the sealing part is brazed to the shielding insert.
6. The system of claim 4, further comprising a spring disposed between the sealing part and the shielding insert.
7. The system of claim 1, further comprising a shielding connection block configured to couple the shielding insert to the chassis shielding block.
8. The system of claim 7, wherein the shielding connection block is made of dense shielding material.
9. The system of claim 7, further comprising a threaded fastener configured to couple together the shielding insert and the shielding connection block.
10. The system of claim 7, further comprising the nuclear logging tool comprising the collar disposed about the chassis, wherein a shielding port extends radially through the collar toward the chassis, the shielding insert is disposed in the shielding port over the chassis shielding block, a sealing part is disposed in the shielding port over the shielding insert, and the shielding top plate is coupled to the collar over the sealing part.
11. The system of claim 10, wherein the shielding port comprises an annular sealing port having a first annular bore, a second annular bore, and an annular ledge between the first and second annular bores, wherein the sealing part comprises an annular sealing cap having a cup-shaped structure disposed in the first annular bore, wherein the first portion of the shielding insert comprises a first annular portion disposed in the cup-shaped structure, the second portion of the shielding insert comprises a second annular portion disposed in the second annular bore, and an annular flange of the shielding insert is disposed between the cup-shaped structure and the annular ledge.
12. The system of claim 11, wherein a first threaded fastener extends through the shielding insert and couples to a second threaded fastener in the shielding connection block, the chassis shielding block, or the chassis, wherein the first and second threaded fasteners are configured to thread together to hold the shielding insert in the shielding port.
13. The system of claim 11, wherein the shielding top plate is coupled to the collar via one or more fasteners, a brazed joint, a welded joint, or an adhesive joint.
14. The system of claim 1, further comprising a threaded fastener configured to couple together the shielding insert and the chassis shielding block.
15. A nuclear logging tool, comprising: a chassis; a collar disposed about the chassis; a gamma ray source configured to emit gamma rays through a first window in the collar; a gamma ray detector configured to receive gamma rays through a second window in the collar, wherein the gamma ray source and the gamma ray detector are offset from one another along an axis of the chassis; and a shield comprising: a shielding insert mounted at least partially inside the nuclear logging tool between the gamma ray source and the gamma ray detector, wherein a first portion of the shielding insert is mounted in the collar and a second portion of the shielding insert is mounted in the chassis; a chassis shielding block mounted in the chassis between the gamma ray source and the

gamma ray detector; and a shielding top plate coupled to the collar and at least partially retaining the shielding insert in the collar.

16. The nuclear logging tool of claim 15, wherein a shielding port extends radially through the collar toward the chassis, the shielding insert is disposed in the shielding port over the chassis shielding block, a sealing part is disposed in the shielding port over the shielding insert, and the shielding top plate is coupled to the collar over the sealing part.

17. The nuclear logging tool of claim 16, wherein the shielding port comprises an annular sealing port having a first annular bore, a second annular bore, and an annular ledge between the first and second annular bores, wherein the sealing part comprises an annular sealing cap having a cup-shaped structure disposed in the first annular bore, wherein the first portion of the shielding insert comprises a first annular portion disposed in the cup-shaped structure, the second portion of the shielding insert comprises a second annular portion disposed in the second annular bore, and an annular flange of the shielding insert is disposed between the cup-shaped structure and the annular ledge.

18. The nuclear logging tool of claim 17, further comprising a shielding connection block configured to couple the shielding insert to the chassis shielding block, wherein a first threaded fastener extends through the shielding insert and couples to a second threaded fastener in the shielding connection block, the chassis shielding block, or the chassis, wherein the first and second threaded fasteners are configured to thread together to hold the shielding insert in the shielding port.

19. A method, comprising: emitting gamma rays from a gamma ray source through a first window in a collar of a nuclear logging tool; receiving gamma rays at a gamma ray detector through a second window in the collar, wherein the gamma ray source and the gamma ray detector are offset from one another along an axis of a chassis of the nuclear logging tool; and shielding gamma rays directly between the gamma ray source and the gamma ray detector internally within the nuclear logging tool via a shield, wherein the shield comprises: a shielding insert mounted at least partially inside the nuclear logging tool between the gamma ray source and the gamma ray detector, wherein a first portion of the shielding insert is mounted in the collar and a second portion of the shielding insert is mounted in the chassis; a chassis shielding block mounted in the chassis between the gamma ray source and the gamma ray detector; and a shielding top plate coupled to the collar and at least partially retaining the shielding insert in the collar.

20. The method of claim 19, wherein emitting the gamma rays comprises directing the gamma rays through a geological formation, receiving the gamma rays comprises receiving the gamma rays after passing through the geological formation, and shielding the gamma rays comprises reducing noise associated with the gamma rays not passing through the geological formation.